

The Polar Express: Dynamic Aggregation under Policy Holonomy

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September 2026

Abstract

A low-dimensional organizational report can lose sufficiency for future responses after a feasible policy cycle. Policy technology, equilibrium drift, and an implementation metric determine how motion in the report transports the latent state. A report can therefore return to its starting point while the internal state finishes elsewhere. A common future probe measures the resulting target holonomy. Rectangles traversed in opposite directions identify target curvature, while mirrored loops remove the cubic term. The row space of this curvature determines which target defects can be closed through loop design. Pulling target covectors back through the loop library gives the local state information that must accompany the report. A phase-conditional Gorman form attains this dimension when the transported span has an additive differential frame. The construction is studied in a 72-state transition economy and in public LaDe delivery records. In the transition economy, a closed output–consumption loop changes a future consumption response by up to 6.04 percent of its ordinary value. In 4,514,658 package records, six additive parcel, courier, and delivery-area moments reduce next-day backlog RMSE by 6.13 percent after conditioning on report history and accepted flow. The LaDe exercise measures predictive closure under observed operations. Assigned report loops and terminal probes would identify response holonomy.

JEL codes: C14, C32, C61, C93, D12, E47, L91. *Keywords:* dynamic aggregation, holonomy, policy experiments, inverse design, Gorman aggregation, heterogeneous agents, Express operations.

1. Introduction

The reported organization has returned. Whether its constituents have returned is an empirical question. An Express network can return throughput, mean delivery time, backlog, and active capacity to their initial values after a sequence of routing, incentive, and capacity choices. Parcel ages, courier loads, and delivery-area congestion can finish elsewhere. The same demand-surge or temporary-capacity probe can then generate a different clearance path. Headquarters sees the same dashboard while future operations reveal another state.

Gorman's aggregation question changes when the report travels through a policy history. A large organization summarizes heterogeneous parcels, workers, facilities, inventories, and local markets in a small report. Gorman's polar form uses additive micro coordinates and common target slopes. The coordinates that must accompany the report depend on the family of future probes and the policy loops under consideration.

The empirical analysis begins by asking whether additive moments of parcels, couriers, and delivery areas predict future clearance after conditioning on the dashboard and its history. It then describes the loop orientations and terminal probe that would turn this predictive exercise into a causal holonomy experiment. The transition economy provides the corresponding equilibrium calculation in an environment where the full state and the policy lift are observed.

Let $q : \mathcal{S} \rightarrow \mathcal{X}$ map the economic state into an m -dimensional report space. The fiber $q^{-1}(x)$ contains latent distributions, expectations, network states, and algorithmic states that share the report x . Let $\tau(s) \in \mathbb{R}^r$ denote the expected response to a standardized future probe. For a feasible report loop γ based at $x = q(s)$ whose lift exists from s , define the target holonomy defect

$$(1) \quad h_\tau(\gamma, s) = \tau(\text{Hol}_\gamma(s)) - \tau(s).$$

The report retains the state information used by the probe along policy cycles when $h_\tau(\gamma, s) = 0$ throughout the accessible holonomy orbit. Sufficiency is therefore specific to the report, policy technology, implementation rule, and target. The same displacement along a fiber may be invisible to one probe and visible to another.

Equation (1) organizes the paper. Policy transport defines Hol_γ . Reversing the direction of a loop identifies the target derivative of that transport. Oriented areas provide coordinates for constructing a closing loop. Finally, Gorman coordinates describe the state information needed to preserve the future response across the chosen loop library.

Figure 1 displays these objects in the transition economy used below. The two orientations trace the same closed output-consumption report path. Their terminal distributions differ on the asset-income fiber. Applying the common future consumption probe con-

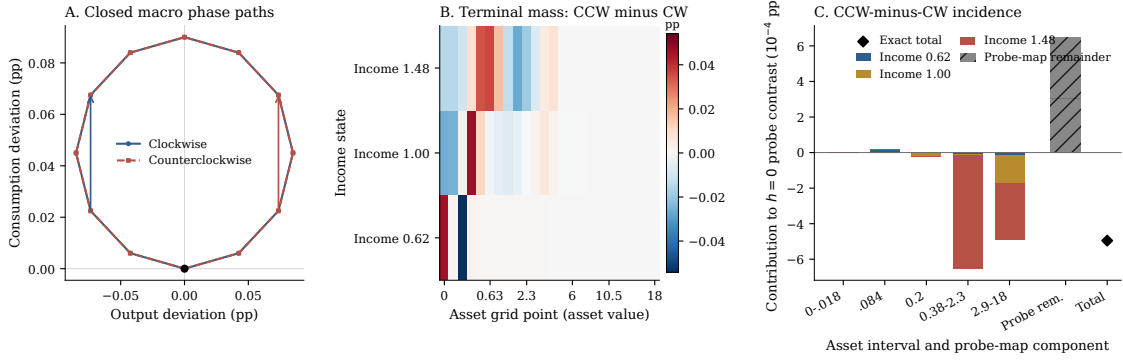


FIGURE 1. A closed report path, transported state, and target defect

Panel A plots the clockwise and counterclockwise output–consumption paths in the 72-state transition economy. Both begin and end at the black report point. Panel B gives the percentage-point difference between the two terminal asset–income distributions. Panel C decomposes the horizon-0 consumption–probe contrast into asset–income contributions and the probe-map remainder. The underlying transition paths, terminal distributions, and probe weights generate every plotted coordinate.

verts that distributional displacement into the target contrast shown in panel C.

The controlled law $\dot{s} = f_s + G_s u$ induces the policy-to-report map $A_s = Dq_s G_s$. Its rank ρ may be smaller than the report dimension m , since the available instruments need only move the report along a subset of directions. On a constant-rank domain, these attainable directions form a distribution $\mathcal{E} \subseteq T\mathcal{X}$. A quadratic implementation criterion selects the Moore–Penrose lift of each velocity in $\mathcal{E}$. This partial connection determines which report paths are feasible, how each path transports the state, and which map of the observational fiber is generated when the report path closes.

The inverse-design idea is related to a trajectoid. A rolling constraint lifts a planar path into the orientation of a body, and the accumulated orientation is the holonomy of that lift. The shape of the body can then be chosen so that its orientation reaches a prescribed stabilizer (Sobolev et al. 2023). In the economic setting, instruments, equilibrium restrictions, drift, and implementation cost select the lift, while the hidden coordinate is the internal state of the organization. The geometry of the report loop is chosen so that the future probe response reaches its stabilizer. For a small loop, oriented area governs the leading hidden displacement. When instantaneous policy access has lower dimension, successive commutators generate further directions, as in locomotion under rolling constraints (Bloch et al. 1996).

On an involutive accessible leaf where the autonomous drift is horizontal under the selected lift, let $v, w \in \mathcal{E}_x$, let $\Omega_s(v, w)$ be the vertical curvature of the policy lift, and let γ_ε^+ and γ_ε^- be the two orientations of an ε -rectangle. The direction-reversal theorem gives

$$(2) \quad \frac{\tau(\text{Hol}_{\gamma_\varepsilon^+}(s)) - \tau(\text{Hol}_{\gamma_\varepsilon^-}(s))}{2\varepsilon^2} \longrightarrow D\tau_s \Omega_s(v, w).$$

The empirical contrast uses identical report rectangles traversed in opposite directions, followed by the same terminal probe. Four mirrored arms remove the cubic term. With J independent clusters assigned to policy paths, the estimator has bias $O(\varepsilon^2)$ and variance $O((J\varepsilon^4)^{-1})$. A matched-set design forms the four-arm contrast within each set and uses independence across sets for inference. If each cluster contains M terminal observations, the variance of an arm mean can be written as $\Sigma_B/J + \Sigma_W/(JM)$.

This assignment structure allows users and sellers to interact within a market while maintaining path-level independence across clusters. The clusters may be economies, jurisdictions, markets, or other policy units. An exogenous environment can also enter the lift, yielding a variational formula for changes in the defect across environments.

The terminal probe is supplied by an independently randomized shock. Closure at each prespecified shock level depends on the transported level-response covectors. First-order robustness to the shock index additionally depends on their derivatives with respect to that shock. The two spans may have different dimensions, so a report can preserve the response at the chosen shock levels while leaving its local comparative statics unresolved.

Curvature also gives coordinates for inverse design. The attainable distribution provides $L = \dim \wedge^2 \mathcal{E}_x = \binom{\rho}{2}$ elementary oriented areas. After a basis is chosen, their effects on the target form a matrix $K_s \in \mathbb{R}^{L \times r}$. A baseline target defect b admits a second-order area bridge when $b \in \text{row}(K_s)$, and

$$(3) \quad a^* = -bK_s^\dagger$$

selects the minimum-norm vector of oriented areas. Reaching every local target requires $\text{rank}(K_s) = r$, which implies $\binom{\rho}{2} \geq r$. Rectangles with side lengths proportional to $\sqrt{|a_j|}$ implement the selected areas. Their target remainder is $O(\|a\|^{3/2})$, so the bridge can be updated as the residual falls. Under noninvolutive access, commutator words of length ℓ replace rectangles as the local building blocks. Their signed coefficients have remainder $O(|a|^{1+1/\ell})$, and their target images replace the area rows in the bridge equation.

The same loop library determines which state information must travel with the report. Linearizing the report and probe response gives maps O and D . The rank of D on $\ker O$ is the number of target-sensitive measurement coordinates required for linear reconstruction. This connects the endpoint problem to functional observability and goal-oriented sensor design (Montanari et al. 2022; Wu et al. 2023). In applications, weighted micro moments can recover these coordinates from sampled parcels, couriers, households, users, or sellers.

Gorman's polar form provides an aggregation rule for such moments. The phase-conditional form is $\tau(s) = \alpha(q(s)) + M(s)B(q(s))$, where M contains additive moments and B is a common target slope (Gorman 1953, 1961). Pulling each endpoint target covector back through the differential of its holonomy map gives a span in the initial observational

fiber. Its dimension is $d_G^{\text{loc}}(s)$. A measurement is first-order sufficient for the loop library when its fiber differentials contain this span. An additive Gorman representation reaches the lower bound when the span has an additive differential frame.

Finite loop histories require a corresponding transition rule. For each loop type j , the retained moments must satisfy the affine closure relation $m^+ = h_x A_j + m R_j$. In the finite linear model, the transported covector span is the common invariant subspace generated by the probe kernel after phase-known loadings are removed. A finite iteration constructs this quotient.

Repeated policy episodes can be used to estimate (A_j, R_j, B) . For a fixed loop word, least-squares estimators give a root- J plug-in limit at the policy-cluster level. Matrix products and common slopes then transport retained moments through a finite loop history, while local direction reversal estimates the area derivative. The two calculations provide a local diagnostic of hidden displacement and a transition equation for recurrent policy histories.

Connections, holonomy, and curvature are used in geometric mechanics and locomotion (Ambrose and Singer 1953; Kobayashi and Nomizu 1963; Kelly and Murray 1995; Bloch et al. 1996). Controllability and observability organize controlled systems (Hermann and Krener 1977), while econometric identification studies observational equivalence in latent dynamics (Rothenberg 1971; Hu and Shum 2012). Exact aggregation and lumpability concern finite representations of a larger state (Gorman 1953, 1961; Buchholz 1994). Marketplace experiments address interference through shared supply and demand (Johari et al. 2022; Holtz et al. 2025; Bajari et al. 2023), and performative prediction studies feedback from deployed algorithms to the data-generating state (Perdomo et al. 2020). Here policy technology generates the connection on an observational fiber, a future probe defines the estimand, and an oriented report loop supplies the intervention.

Express operations and a 72-state perfect-foresight transition equilibrium provide the two implementations. The LaDe sample contains 4,514,658 package records and 30,534 region-days. The operational report records throughput, mean lead time, backlog, and active couriers. Six additive moments retain information about backlog age and concentration across couriers and delivery areas. On within-region terminal holdouts, adding these moments lowers next-day backlog RMSE by 6.13 percent after conditioning on current reports, exact-calendar lags, and accepted-parcel flow. Squared risk falls by 11.89 percent, with a region-and-date clustered t statistic of 2.94. Absolute risk falls by 8.83 percent, with $t = 6.17$. The absolute-loss gain is positive in every city and under each workload restriction. Rolling-origin evaluation and conditioning on common future arrivals provide checks across time and input paths.

These historical records identify predictive closure under observed operations. The causal design would assign report report loops and a common terminal probe. In the transi-

tion economy, three policy coordinates in each of 12 quarters form a 36-dimensional policy path, while the requested output–consumption trace imposes 24 equilibrium restrictions. The computed path Jacobian has rank 24. Across the loops considered, the counterclockwise horizon-0 defect ranges from 0.36 to 6.04 percent of the ordinary probe response.

Policy transport leads from the target-holonomy criterion to the direction-reversal experiment and curvature-based inverse design. The same loop library determines the minimum Gorman state and its finite invariant quotient. The transition equilibrium and Express operations then separate observed predictive closure from the assigned-loop protocol required for causal holonomy.

2. Policy transport and target holonomy

A reported path moves latent state through the policy instruments used to implement it. The transport first appears in state space and then on policy-path space. Closed paths generate fiber maps whose curvature and higher Lie words determine whether the probe target returns to its initial value.

2.1. Economic state, report, and attainable phase motion

Let $\mathcal{S}$ be an n -dimensional smooth state manifold and let $\mathcal{X}$ be an m -dimensional smooth phase manifold equipped with a Riemannian metric. The map $q : \mathcal{S} \rightarrow \mathcal{X}$ reports the phase point. Its vertical space at s is

$$(4) \quad \mathcal{V}_s = \ker Dq_s = T_s q^{-1}(q(s)).$$

The controlled state law is

$$(5) \quad \dot{s} = f_s + G_s u, \quad d_s = Dq_s f_s, \quad A_s = Dq_s G_s : \mathbb{R}^k \rightarrow T_{q(s)} \mathcal{X},$$

where f is the autonomous drift, $u \in \mathbb{R}^k$ is the policy deviation, and R_s is the positive-definite implementation-cost matrix. A prime on a map into a phase tangent denotes the adjoint under the Euclidean policy inner product and the phase metric. The attainable relative phase velocities form $\text{im } A_s$.

ASSUMPTION 1 (Constant-rank policy technology). *The maps q , f , G , and R are C^3 , q is a submersion, R_s is symmetric positive definite, and A_s has constant rank $\rho \leq m$. There are a smooth rank- ρ subbundle $\mathcal{E} \subseteq T\mathcal{X}$ and a vector field d on $\mathcal{X}$ such that*

$$(6) \quad \text{im } A_s = \mathcal{E}_{q(s)}, \quad Dq_s f_s = d_{q(s)}.$$

The projectability conditions require states with the same report to have the same set of attainable phase velocities. This restriction can be assessed by comparing policy-to-phase maps within an observational fiber. A requested phase velocity $v \in T_x\mathcal{X}$ at $x = q(s)$ is feasible when $e = v - d_x \in \mathcal{E}_x$. Among the policies that deliver this velocity, select

$$(7) \quad u^E(s, v) = \arg \min_{u \in \mathbb{R}^k} \frac{1}{2} u' R_s u \quad \text{subject to} \quad A_s u = v - d_x.$$

Write $C_s = A_s R_s^{-1} A_s'$ and let $C_s^\dagger$ be its Moore–Penrose inverse. Define the relative state lift $L_s^E : \mathcal{E}_x \rightarrow T_s\mathcal{S}$ by

$$(8) \quad u^E(s, v) = R_s^{-1} A_s' C_s^\dagger (v - d_x), \quad L_s^E e = G_s R_s^{-1} A_s' C_s^\dagger e.$$

PROPOSITION 1 (Partial policy connection). *Under assumption 1, equation (8) is the unique solution of (7). For every $e \in \mathcal{E}_x$, it satisfies $Dq_s L_s^E e = e$ and determines the decomposition*

$$(9) \quad Dq_s^{-1}(\mathcal{E}_x) = \mathcal{V}_s \oplus \mathcal{H}_s^E, \quad \mathcal{H}_s^E = \text{im } L_s^E.$$

If $\mathcal{E}$ is involutive, $\mathcal{H}^E$ is an Ehresmann connection on the restriction of the reported-state bundle to each integral leaf of $\mathcal{E}$. If $\rho = m$, equation (9) becomes $T_s\mathcal{S} = \mathcal{V}_s \oplus \mathcal{H}_s^E$ and $C_s^\dagger = C_s^{-1}$.

The instrument set, equilibrium restrictions, drift compensation, and R_s jointly determine the partial connection. Changing one of these objects changes L_s^E and can alter the state displacement accumulated along a feasible phase loop. Where rank changes, the state space separates into regular strata. Within a stratum, the smallest positive singular value of $A_s R_s^{-1/2}$ records the local conditioning of implementation.

When $d_x \in \mathcal{E}_x$, decompose the autonomous drift as

$$(10) \quad f_s = L_s^E d_x + b_s, \quad b_s \in \mathcal{V}_s.$$

The field b_s is the vertical residual drift relative to the selected minimum-cost lift. On a domain where $b_s = 0$, equation (12) reduces to $\dot{s}_t = L_{s_t}^E \dot{\gamma}_t$. Spatial rectangles are then flow commutators of the relative horizontal lifts, which is the setting used for the local curvature results below. A controller can produce the same reduction by offsetting b_s and recentering the policy. With a general b_s , finite-time fiber transport also depends on loop duration; the path-space and polar-transport objects retain this dependence.

2.2. A connection on policy-path space

In a forward-looking equilibrium, current policy is linked to the entire report path. Fix an initial state and a terminal condition, choose a local chart for each reported date, and stack T policy vectors as $U \in \mathbb{R}^{kT}$. Let $\Phi_T(U; s_0) \in \mathbb{R}^{mT}$ denote the resulting equilibrium report path, with the stacked coordinates carrying the product phase metric. A requested path $\gamma \in \mathbb{R}^{mT}$ is locally feasible at U_0 when $\gamma - \Phi_T(U_0; s_0)$ belongs to the range of $D_U \Phi_T(U_0; s_0)$. Given a positive-definite path cost R_T , the linearized lift is

$$(11) \quad \Delta U^H = R_T^{-1} D\Phi_T'(D\Phi_T R_T^{-1} D\Phi_T')^\dagger (\gamma - \Phi_T(U_0; s_0)).$$

PROPOSITION 2 (Path-space policy connection). *Suppose Φ_T is C^1 and $D_U \Phi_T$ has constant rank ρ_T near U_0 . Equation (11) is the minimum- R_T -norm solution of the linearized path restriction. If $\rho_T = mT$, the exact solution set $\{U : \Phi_T(U; s_0) = \gamma\}$ is a local manifold of dimension $kT - mT$ around each nearby solution, and the Newton update based on (11) is locally defined. If $\rho_T < mT$, the same expression lifts the attainable projection of the requested path and leaves an orthogonal feasibility residual.*

The path-space map imposes expectations, market clearing, and terminal conditions before it is differentiated. Its derivative therefore supplies the policy connection for a perfect-foresight or sequence-space implementation. Singular values recomputed along the Newton iterates show whether the solution remains in the same regular stratum and how well conditioned the requested loop is.

2.3. Accessible paths and target holonomy

A C^1 phase path $\gamma : [0, 1] \rightarrow \mathcal{X}$ is policy-feasible when $\dot{\gamma}_t - d_{\gamma(t)} \in \mathcal{E}_{\gamma(t)}$. For an initial state s satisfying $q(s) = \gamma(0)$, its lift solves

$$(12) \quad \dot{s}_t = f_{s_t} + L_{s_t}^E(\dot{\gamma}_t - d_{\gamma(t)}), \quad s_0 = s.$$

Equations (6) and (9) give $Dq_{s_t} \dot{s}_t = \dot{\gamma}_t$, hence $q(s_t) = \gamma(t)$ throughout the lift. The analysis uses feasible paths for which this solution stays in the constant-rank domain. If $\gamma(1) = \gamma(0) = x$, let $\mathcal{D}_\gamma \subseteq q^{-1}(x)$ be the open set of initial states whose lifted solutions exist through time one and remain in the constant-rank domain. The endpoint defines a local accessible fiber map

$$(13) \quad \text{Hol}_\gamma : \mathcal{D}_\gamma \rightarrow q^{-1}(x), \quad \text{Hol}_\gamma(s) = s_1.$$

The drift and local sections of $\mathcal{E}$ generate control orbits containing the feasible phase paths. Closed paths within an orbit generate a local semigroup of partially defined fiber

maps. If the negative of every feasible lifted field is also feasible, reversed paths provide local inverses, and the maps form a local pseudogroup on their common domains. This formulation allows $\rho < m$ and accommodates a noninvolutive $\mathcal{E}$, for which iterated brackets can enlarge the reachable phase set.

Let a standardized future probe policy be fixed. The target $\tau : \mathcal{S} \rightarrow \mathbb{R}^r$ is the conditional mean of its recorded response, which can contain several response horizons. The target holonomy defect is equation (1).

DEFINITION 1 (Loop sufficiency). *Let $\mathcal{O}_s^{\text{Hol},E}$ be the set reached from s by lifts of feasible contractible loops based at $q(s)$. The phase report is accessibly loop-sufficient for τ at s when τ is constant on $\mathcal{O}_s^{\text{Hol},E}$.*

PROPOSITION 3 (Holonomy-orbit criterion). *Suppose $\mathcal{O}_s^{\text{Hol},E}$ is a connected immersed submanifold. The phase report is accessibly loop-sufficient for τ at s exactly when*

$$(14) \quad D\tau_z \xi = 0 \quad \text{for every } z \in \mathcal{O}_s^{\text{Hol},E} \text{ and } \xi \in T_z \mathcal{O}_s^{\text{Hol},E}.$$

If the accessible holonomy orbit equals the connected component of the fiber, this condition is equivalent to constancy of τ on that fiber component. When parallel transports of feasible curvature and commutator values span the accessible holonomy-orbit tangent space, equation (14) is equivalent to annihilation of those transported values.

2.4. Exogenous environments and comparative statics

Let the scalar η index an exogenous environment such as demand intensity, weather severity, input cost, or available capacity. Keep the report map q fixed and allow $(f^\eta, G^\eta, R^\eta, \tau^\eta)$, the attainable distribution $\mathcal{E}^\eta$, and its lift $L^{E,\eta}$ to vary smoothly with η . Fix a report loop γ that remains feasible on a neighborhood of η_0 and define

$$(15) \quad b_t^\eta(z) = f_z^\eta + L_z^{E,\eta}(\dot{\gamma}_t - d_{\gamma(t)}^\eta), \quad \dot{S}_t^\eta = b_t^\eta(S_t^\eta), \quad S_0^\eta = s_\eta^\circ.$$

The endpoint is $\text{Hol}_\gamma^\eta(s_\eta^\circ) = S_1^\eta$, and the shock-indexed target defect is $h_t^\eta(\gamma, s_\eta^\circ) = \tau^\eta(S_1^\eta) - \tau^\eta(s_\eta^\circ)$. The baseline s_η° may be a shock-specific stationary state or a common initial state.

PROPOSITION 4 (Shock-indexed holonomy comparative statics). *Suppose b_t^η is continuously differentiable in (z, η) , τ^η and s_η° are continuously differentiable in η , and the lifted solutions remain in a common compact regularity neighborhood. Then $J_t^\eta = \partial_\eta S_t^\eta$ solves*

$$(16) \quad \dot{J}_t^\eta = D_s b_t^\eta(S_t^\eta) J_t^\eta + \partial_\eta b_t^\eta(S_t^\eta), \quad J_0^\eta = \partial_\eta s_\eta^\circ,$$

and

$$(17) \quad \begin{aligned} \frac{d}{d\eta} h_{\tau}^{\eta}(\gamma, s_{\eta}^{\circ}) &= \partial_{\eta} \tau^{\eta}(S_1^{\eta}) + D_s \tau^{\eta}(S_1^{\eta}) J_1^{\eta} \\ &\quad - \partial_{\eta} \tau^{\eta}(s_{\eta}^{\circ}) - D_s \tau^{\eta}(s_{\eta}^{\circ}) \partial_{\eta} s_{\eta}^{\circ}. \end{aligned}$$

At a fixed base state and fixed phase directions, $\kappa_{\tau}^{\eta} = D_s \tau^{\eta} \Omega^{\eta}$ satisfies $\partial_{\eta} \kappa_{\tau}^{\eta} = D_s(\partial_{\eta} \tau^{\eta}) \Omega^{\eta} + D_s \tau^{\eta} \partial_{\eta} \Omega^{\eta}$.

Equation (17) separates a direct change in the probe mapping from the change in state transported by the shock-indexed policy connection. For a stochastic innovation path, the equation applies conditionally on the realized path. Integrating over a prespecified law gives the regime-specific expected defect, and a path experiment balances this law across orientation arms through randomization within calendar and forecast blocks. Two fixed shock levels identify a finite change; dividing by their intensity gap gives a secant. With symmetric levels $\eta_0 \pm \delta$, the derivative estimator is

$$(18) \quad \widehat{\partial_{\eta} \kappa}_{\varepsilon, \delta} = \frac{\widehat{\kappa}_{\varepsilon}^{\eta_0 + \delta} - \widehat{\kappa}_{\varepsilon}^{\eta_0 - \delta}}{2\delta}.$$

Under a symmetric four-arm path design at each shock level, uniform fourth-order path expansions, three derivatives in η , and balanced independent clusters, its bias is $O(\varepsilon^2 + \delta^2)$ and its variance is $O((J\varepsilon^4\delta^2)^{-1})$. Estimating an infinitesimal comparative static therefore requires both the path and shock increments to shrink. A fixed pair of shock levels instead yields the corresponding finite change.

2.5. An area law in three coordinates

The following three-coordinate example makes the area calculation explicit. Let $\mathcal{S} = \mathbb{R}^3$, with state $s = (x, y, h)$ and report $q(s) = (x, y)$. The horizontal vector fields associated with the two phase coordinates are

$$(19) \quad H_1 = \partial_x, \quad H_2 = \partial_y + x\partial_h.$$

A move of length a along H_1 changes x . A move of length b along H_2 changes y and adds xb to h . Compose the moves $+aH_1, +bH_2, -aH_1, -bH_2$.

PROPOSITION 5 (Policy rectangle). *The rectangle returns (x, y) to its initial value and maps h to $h + ab$. Reversing its orientation maps h to $h - ab$. For a linear probe $\tau(s) = h\ell \in \mathbb{R}^r$,*

$$(20) \quad \tau(\text{Hol}_{\gamma_{a,b}^+}(s)) - \tau(\text{Hol}_{\gamma_{a,b}^-}(s)) = 2abl.$$

An additive correction $c = -ab$ closes the scalar target, and two opposite rectangles close it by

composition.

The Lie bracket $[H_1, H_2] = \partial_h$ is vertical, and ab is the signed phase area. Proposition 5 supplies an exact discrete version of the small-loop calculation that follows.

2.6. Target curvature and phase sufficiency

Curvature on an accessible leaf. Suppose first that $\mathcal{E}$ is involutive near x , and let $\mathcal{L}_x$ be its ρ -dimensional integral leaf. On $Dq_s^{-1}(\mathcal{E}_x)$, let $P_s^{\mathcal{V},E} = I - L_s^E Dq_s$ be the vertical projection from (9). For $v, w \in T_x \mathcal{L}_x = \mathcal{E}_x$, extend them to commuting coordinate vector fields on the leaf and let v^H, w^H denote their relative horizontal lifts. Define

$$(21) \quad \Omega_s^E(v, w) = P_s^{\mathcal{V},E}[v^H, w^H]_s \in \mathcal{V}_s, \quad \kappa_\tau^E(s; v, w) = D\tau_s \Omega_s^E(v, w) \in \mathbb{R}^r.$$

The first object is the curvature of the partial policy connection under the stated sign convention. The second retains the component read by the future probe. Subsequent formulas suppress the superscript E .

Let the two coordinate rectangles have ordered sides

$$(22) \quad \gamma_\varepsilon^+ : (+\varepsilon v, +\varepsilon w, -\varepsilon v, -\varepsilon w), \quad \gamma_\varepsilon^- : (+\varepsilon w, +\varepsilon v, -\varepsilon w, -\varepsilon v).$$

Both loops start and finish at x .

THEOREM 1 (Direction reversal and target curvature). *Suppose the accessible-leaf connection and τ are C^4 , $d \in \mathcal{E}$, and the vertical residual drift in (10) vanishes on the flow neighborhood. In a state chart around s ,*

$$(23) \quad \text{Hol}_{\gamma_\varepsilon^+}(s) = s + \varepsilon^2 \Omega_s(v, w) + O(\varepsilon^3),$$

$$(24) \quad \text{Hol}_{\gamma_\varepsilon^-}(s) = s - \varepsilon^2 \Omega_s(v, w) + O(\varepsilon^3),$$

and

$$(25) \quad \frac{\tau(\text{Hol}_{\gamma_\varepsilon^+}(s)) - \tau(\text{Hol}_{\gamma_\varepsilon^-}(s))}{2\varepsilon^2} = \kappa_\tau(s; v, w) + O(\varepsilon).$$

Let $\Delta(\varepsilon)$ denote the numerator in (25), and construct mirrored loops by replacing ε with $-\varepsilon$ in every side. Then

$$(26) \quad \frac{\Delta(\varepsilon) + \Delta(-\varepsilon)}{4\varepsilon^2} = \kappa_\tau(s; v, w) + O(\varepsilon^2).$$

The two-loop contrast identifies the sign associated with oriented area. The third-order

term changes sign under $\varepsilon \mapsto -\varepsilon$, so the mirror average removes it. In Proposition 5, the remainder is zero and equation (26) holds for every a, b in the stated domain.

COROLLARY 1 (Curvature diagnostic). *If $\kappa_\tau(s; v, w) \neq 0$, the phase report is locally deficient for the probe target along the policy connection. If the transported curvature values span the holonomy-orbit tangent space and $D\tau$ annihilates every one of them, the phase report is loop-sufficient on that orbit.*

The diagnostic is specific to the chosen probe response. A distribution may move after a loop in a direction contained in $\ker D\tau_s$, yielding zero target curvature for that probe. A different probe can load on the same movement and produce a nonzero contrast.

Noninvolutive access and Lie-word closure. When $\mathcal{E}$ is noninvolutive, feasible phase paths lie on the control orbit generated by its local sections. Let $e_1, \dots, e_\rho$ be a local frame, let $H_i = L^E e_i$ be the relative state lifts, and let $\text{Lie}_{\leq \ell}(H)$ denote bracket words of length at most ℓ . The target-relevant vertical word space at s is

$$(27) \quad \mathcal{W}_{\ell, \tau}(s) = \text{span} \{ D\tau_s Z_s : Z \in \text{Lie}_{\leq \ell}(H), Dq_s Z_s = 0 \} \subseteq \mathbb{R}^r.$$

PROPOSITION 6 (Accessible Lie-word bridge). *Consider a nonempty finite reversible library of commutator controls with leading vertical words $Z_1, \dots, Z_J$ of lengths $\ell_1, \dots, \ell_J$, and write $\ell_{\max} = \max_j \ell_j$. Suppose the lifted fields and target have bounded derivatives through order $\ell_{\max} + 1$ on a compact flow neighborhood. Suppose each projected commutator endpoint admits a feasible phase-closing segment whose state contribution has the next asymptotic order. Parameterize the j th signed word by its leading coefficient a_j . Then its target displacement is*

$$(28) \quad h_{\tau, j}(a_j, s) = a_j D\tau_s Z_j(s) + O(|a_j|^{1+1/\ell_j}).$$

The bridgeable baseline defects at leading order form

$$(29) \quad \text{span}\{D\tau_s Z_j(s) : j = 1, \dots, J\}.$$

Full local target reachability requires this span to have dimension r .

The curvature bridge is the length-two case: Z_j is a vertical commutator and a_j is oriented area. Higher words provide additional closure coordinates when instantaneous phase rank is limited and policy sequencing expands the accessible Lie algebra. The corresponding control-system example is a rolling constraint. Its admissible velocities have lower dimension than the configuration space, while repeated commutators still generate displacement (Bloch et al. 1996).

3. Identification and inverse design

The oriented-loop contrast connects the empirical and design parts of the analysis. Random assignment identifies how state transport changes the target. The estimated derivative can then be used as a coordinate system for constructing a loop that offsets an observed target defect.

3.1. Four randomized arms

Index loop orientation by $o \in \{+, -\}$ and mirror sign by $\sigma \in \{+, -\}$. Arm (o, σ) receives the corresponding loop with signed side scale $\sigma\epsilon$, followed by the common probe. Let $Y^P \in \mathbb{R}^r$ be the recorded probe response and let

$$(30) \quad m_{o,\sigma}(\epsilon) = \mathbb{E}[Y^P \mid o, \sigma], \quad \Delta_\sigma(\epsilon) = m_{+,\sigma}(\epsilon) - m_{-,\sigma}(\epsilon).$$

Random assignment identifies these four conditional means. Because the initial-state distribution is shared across arms, their differences isolate the assigned paths. Conditioning and covariate adjustment can be incorporated in the arm means when they are prespecified.

Each assigned path is centered at the unit's initial report $q(S)$ and uses the same local direction fields across arms. Potential-outcome consistency for the probe means

$$(31) \quad \mathbb{E}[Y^P(o, \sigma) \mid S = s] = \tau(\text{Hol}_{\gamma_{\sigma\epsilon, s}^o}(s)),$$

where $\gamma_{\sigma\epsilon, s}^o$ is the corresponding report loop based at $q(s)$. Thus the randomized arm contrast averages terminal-state probe responses under a common initial-state law.

Define the population and sample curvature contrasts

$$(32) \quad \kappa_\epsilon = \frac{\Delta_+(\epsilon) + \Delta_-(\epsilon)}{4\epsilon^2}, \quad \widehat{\kappa}_\epsilon = \frac{\bar{Y}_{+,+} - \bar{Y}_{-,+} + \bar{Y}_{+,-} - \bar{Y}_{-,-}}{4\epsilon^2}.$$

THEOREM 2 (Shrinking-loop identification and inference). *Let $\epsilon = \epsilon_n > 0$ satisfy $\epsilon_n \rightarrow 0$. Suppose independent experimental units are randomized across the four arms with $n_{o,\sigma}/n \rightarrow \pi_{o,\sigma} \in (0, 1)$, equation (31) holds, the arm outcomes have uniformly bounded $2 + \delta$ moments along the shrinking sequence, their covariance matrices converge to $\Sigma_{o,\sigma}$, and the smooth expansion in theorem 1 holds uniformly over the initial-state support. Then*

$$(33) \quad \kappa_\epsilon = \mathbb{E}[\kappa_\tau(S; \nu, w)] + O(\epsilon^2).$$

The area-normalized sampling limit is

$$(34) \quad \sqrt{n}\varepsilon^2(\widehat{\kappa}_\varepsilon - \kappa_\varepsilon) \implies \mathcal{N}(0, V), \quad V = \frac{1}{16} \sum_{o,\sigma} \frac{\Sigma_{o,\sigma}}{\pi_{o,\sigma}}.$$

Consistency follows when $n\varepsilon^4 \rightarrow \infty$. Centering the limit at $\mathbb{E}[\kappa_\tau(S; \nu, w)]$ additionally follows when $\sqrt{n}\varepsilon^4 \rightarrow 0$. For $\varepsilon = n^{-a}$, both conditions hold when $1/8 < a < 1/4$.

In a market experiment, the independent units of assignment are policy clusters indexed by c , with $A_c = (o, \sigma)$ denoting the assigned path. The potential cluster outcome $\bar{Y}_c^p(a)$ averages terminal probe outcomes within cluster c under path a and allows users and sellers in that cluster to interact. Partial interference requires this outcome to depend on the cluster's assignment and within-cluster exposure, while assignments to other clusters leave it unchanged.

THEOREM 3 (Cluster-assigned path identification). *Suppose policy clusters are independently and identically sampled and paths are randomized with probabilities $\pi_a \in (0, 1)$. Suppose partial interference holds at the chosen cluster partition and the arm-specific cluster means have uniformly bounded $2 + \delta$ moments along the sampling sequence. Then each arm mean identifies $m_a = \mathbb{E}[\bar{Y}_c^p(a)]$, the clockwise-counter-clockwise contrast identifies $m_{+,+} - m_{-,+}$, and the four-arm contrast in (32) identifies κ_ε . If arm a contains J_a clusters and M_a terminal observations per cluster, with*

$$(35) \quad \text{Var}(\bar{Y}_c^p(a)) = \Sigma_{B,a} + \frac{\Sigma_{W,a}}{M_a},$$

then, conditional on the realized arm count J_a or under fixed-count randomization,

$$(36) \quad \text{Var}(\bar{Y}_a) = \frac{\Sigma_{B,a}}{J_a} + \frac{\Sigma_{W,a}}{J_a M_a}.$$

Along a sequence with $J = \sum_a J_a \rightarrow \infty$, $J_a/J \rightarrow \pi_a$, and convergent arm-specific cluster-mean covariance matrices, the cluster central limit theorem gives the finite-loop and area-normalized limits after substituting (36) into their arm contrasts. Cluster-robust arm covariances remain valid when within-cluster dependence is summarized directly by $\text{Var}(\bar{Y}_c^p(a))$.

3.2. Randomized shocks and response holonomy

Shock timing distinguishes two estimands. The first asks whether the completed path changes the response to a common terminal probe. Cluster c completes assigned path a , after which eligible terminal micro units receive shock level $Z_{ic} \in \{0, 1\}$ with prespecified probability p_c . At a fixed saturation and exposure mapping, the Horvitz–Thompson

response score

$$(37) \quad \bar{R}_c(a) = \frac{1}{M_c} \sum_{i=1}^{M_c} \left(\frac{Z_{ic} Y_{ic}}{p_c} - \frac{(1 - Z_{ic}) Y_{ic}}{1 - p_c} \right)$$

identifies the cluster response to the assigned demand, priority, or capacity probe under its randomization design. Substituting $\bar{R}_c(a)$ for $\bar{Y}_c^P(a)$ in equation (32) identifies the target curvature of the shock response $\tau^R(s) = \mathbb{E}[Y(1) - Y(0) \mid S = s]$. This estimand compares the causal response to the same terminal shock after opposite report loops. Because shock assignment begins after the path endpoint is recorded, the endpoint map Hol_γ is held fixed and the contrast isolates the probe-response channel.

The second estimand is the connection comparative static in Proposition 4. Here the environment is assigned before the path and maintained while the path is executed. A randomized capacity envelope, exogenous demand batch, or service constraint can then change both the lifted dynamics and the endpoint map Hol_γ^η . Crossing two environment levels with the orientation-by-mirror assignment gives eight cells and the finite-change estimand

$$(38) \quad \Delta_Z \kappa_\varepsilon = \kappa_\varepsilon^{(1)} - \kappa_\varepsilon^{(0)},$$

where each $\kappa_\varepsilon^{(z)}$ is a four-arm contrast within environment level z . Orthogonal path and environment assignments identify (38). Dividing by the prespecified environment gap gives a secant. With symmetric levels $\eta_0 \pm \delta$, division by 2δ gives equation (18). A cluster-level shock applied after the path yields the same factorial algebra, now with a probe-intensity interpretation and a fixed endpoint map. Weather, upstream flight disruption, and local demand innovations can index observational regimes when their conditional assignment law is justified. In a field design, randomization blocks use preassignment forecasts and calendar waves; realized postassignment environment variables are reserved for reported regime definitions and sensitivity analyses.

THEOREM 4 (Joint shrinking-path and environment inference). *Let $\varepsilon = \varepsilon_J \rightarrow 0$ and $\delta = \delta_J \rightarrow 0$. Suppose independent clusters are randomized across the eight cells $(o, \sigma, z) \in \{+, -\}^3$ with limiting cell shares $\pi_{o,\sigma,z} \in (0, 1)$, uniformly bounded $2 + \nu$ moments, and covariance limits $\Sigma_{o,\sigma,z}$. Suppose uniformly near η_0 ,*

$$(39) \quad \mathbb{E} \widehat{\kappa}_\varepsilon^\eta = \kappa^\eta + \varepsilon^2 B(\eta) + r_\varepsilon(\eta), \quad \sup_\eta \|\partial_\eta r_\varepsilon(\eta)\| = O(\varepsilon^4),$$

where B has a bounded derivative and κ^η has a bounded third derivative. Then the estimator in

equation (18) has bias $O(\varepsilon^2 + \delta^2)$ and

$$(40) \quad \sqrt{J}\varepsilon^2\delta \left(\widehat{\partial_\eta \kappa_{\varepsilon,\delta}} - \mathbb{E}\widehat{\partial_\eta \kappa_{\varepsilon,\delta}} \right) \implies \mathcal{N}(0, V_\eta), \quad V_\eta = \frac{1}{64} \sum_{o,\sigma,z} \frac{\Sigma_{o,\sigma,z}}{\pi_{o,\sigma,z}}.$$

Consistency follows when $J\varepsilon^4\delta^2 \rightarrow \infty$. Centering at $\partial_\eta \kappa^{\eta_0}$ additionally follows when $\sqrt{J}\varepsilon^2\delta(\varepsilon^2 + \delta^2) \rightarrow 0$.

For the digital benchmark, each matched set contains one cluster from every arm. Let $b = 1, \dots, B$ index independent matched sets and let $\bar{Y}_{b,o,\sigma}^p$ be the cluster probe mean in its assigned cell. Define

$$(41) \quad D_b(\varepsilon) = \frac{\bar{Y}_{b,+,+}^p - \bar{Y}_{b,-,+}^p + \bar{Y}_{b,+,-}^p - \bar{Y}_{b,-,-}^p}{4\varepsilon^2}, \quad \kappa_\varepsilon^{\text{MS}} = \mathbb{E}[D_b(\varepsilon)], \quad \widehat{\kappa}_\varepsilon^{\text{MS}} = \frac{1}{B} \sum_{b=1}^B D_b(\varepsilon).$$

THEOREM 5 (Matched-set path inference). *Fix $\varepsilon > 0$. Suppose the matched sets are independently and identically sampled, every set assigns one cluster to each of the four loop cells, path labels are randomized within set, and $\mathbb{E}\|D_b(\varepsilon)\|^{2+\delta} < \infty$. Dependence among the four clusters inside a set is unrestricted. As $B \rightarrow \infty$, $\widehat{\kappa}_\varepsilon^{\text{MS}}$ identifies the average set-level four-arm contrast and*

$$(42) \quad \sqrt{B} \left(\widehat{\kappa}_\varepsilon^{\text{MS}} - \kappa_\varepsilon^{\text{MS}} \right) \implies \mathcal{N}(0, \text{Var}[D_b(\varepsilon)]).$$

The sample variance of $D_b(\varepsilon)$ divided by B estimates the variance of $\widehat{\kappa}_\varepsilon^{\text{MS}}$. Under a sharp null, randomization inference permutes the four path labels within each set.

Squared bias is $O(\varepsilon^4)$ and variance is $O((n\varepsilon^4)^{-1})$. When both leading constants are positive, balancing them gives $\varepsilon \asymp n^{-1/8}$ and root mean squared error of order $n^{-1/4}$. Confidence intervals centered on the limiting curvature use an undersmoothing sequence within the interval in Theorem 2. When $\kappa_\varepsilon = \kappa + c\varepsilon^2 + o(\varepsilon^2)$ uniformly, two loop sizes identify a Richardson extrapolation; its joint arm covariance enters the confidence interval.

3.3. Power frontier and assignment level

The scaling in Theorem 2 also determines which curvature values the experiment can detect. Consider the scalar Gaussian submodel in which arm $a = (o, \sigma)$ supplies n_a independent observations with variance $\sigma_a^2 > 0$ and mean

$$(43) \quad m_a(\varepsilon; \kappa) = \mu_a + s_a \varepsilon^2 \kappa, \quad (s_{+,+}, s_{-,+}, s_{+,-}, s_{-,-}) = (1, -1, 1, -1).$$

The nuisance means μ_a are held fixed across the two simple curvature hypotheses. The same calculation applies when their admissible nuisance subspace is orthogonal to the contrast vector $(s_a)_a$.

THEOREM 6 (Local-holonomy power frontier). *For two curvature values κ_0 and κ_1 , the Kullback–Leibler divergence in the submodel (43) is*

$$(44) \quad \text{KL}(P_{\kappa_0}, P_{\kappa_1}) = \frac{\varepsilon^4(\kappa_1 - \kappa_0)^2}{2} \sum_a \frac{n_a}{\sigma_a^2}.$$

Every level- α test of κ_0 against κ_1 has power bounded above by

$$(45) \quad \alpha + \frac{|\kappa_1 - \kappa_0|\varepsilon^2}{2} \left(\sum_a \frac{n_a}{\sigma_a^2} \right)^{1/2}.$$

Consequently, $n\varepsilon^4(\kappa_1 - \kappa_0)^2 \rightarrow 0$ with stable allocation and variances makes the two sequences asymptotically indistinguishable. For $V > 0$, $\varepsilon > 0$, $\kappa \neq 0$, $\alpha \in (0, 1)$, and $\beta \in (0, 1/2)$, the Wald statistic in (34) has absolute local noncentrality $\sqrt{n}\varepsilon^2|\kappa|/\sqrt{V}$. A sufficient Gaussian planning rule for two-sided size α and power at least $1 - \beta$ takes the smallest integer weakly above the continuous value

$$(46) \quad n_{\alpha, \beta}^{\text{cont}} = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 V}{\varepsilon^4 \kappa^2}.$$

The randomization unit is the economy, jurisdiction, market, or other policy cluster that receives the path. Additional household draws from one terminal distribution reduce measurement error for that realized path, but they do not create independent policy assignments. For a macroeconomic implementation, cluster arm means and cluster-level covariance matrices therefore replace household-level quantities in (34)–(46), with the continuous sample-size value rounded upward. Section 4 develops a finite-loop transport estimator that pools transition information and does not divide the contrast by ε^2 .

3.4. Curvature-rank uncertainty

Experiments on several coordinate planes or at several base points supply the rows of a target-curvature matrix K . Let $\widehat{K}$ be the stacked estimator and suppose a simultaneous confidence construction yields

$$(47) \quad \Pr\{\|\widehat{K} - K\|_{\text{op}} \leq \rho_n\} \geq 1 - \alpha.$$

PROPOSITION 7 (Rank and bridge certificate). *Suppose $L \geq r$. On the event in (47), every singular value satisfies $|\widehat{\sigma}_j - \sigma_j| \leq \rho_n$. Hence $\widehat{\sigma}_r > \rho_n$ certifies full target rank and gives $\sigma_r(K) \geq \widehat{\sigma}_r - \rho_n$. The confidence set*

$$(48) \quad \mathcal{R}_n = \{\text{rank}(K') : \|K' - \widehat{K}\|_{\text{op}} \leq \rho_n\}$$

contains the target-curvature rank with probability at least $1 - \alpha$.

Near a rank change, $\mathcal{R}_n$ can contain several integers, and the area bridge becomes poorly conditioned. Reporting the set makes that uncertainty visible in the design result. Section 3.5 gives the corresponding residual bound.

3.5. Inverse loop design

Area coordinates and accessible phase dimension. Fix $s \in q^{-1}(x)$ on an involutive accessible leaf and choose an orthonormal basis $E_1, \dots, E_L$ of $\Lambda^2 \mathcal{E}_x$ under the metric induced from the phase manifold, where $L = \binom{\rho}{2}$. Stack the feasible target curvatures as rows:

$$(49) \quad K_s = \begin{bmatrix} \kappa_\tau(s; E_1)' \\ \vdots \\ \kappa_\tau(s; E_L)' \end{bmatrix} \in \mathbb{R}^{L \times r}.$$

An area vector $a \in \mathbb{R}^L$ produces the second-order target displacement aK_s . Given a baseline defect $b \in \mathbb{R}^r$, the area closure equation is

$$(50) \quad b + aK_s = 0.$$

THEOREM 7 (Area reachability, codimension, and accessible phase dimension). *Assume $r \geq 1$. Equation (50) has a solution exactly when $b \in \text{row}(K_S)$. On this row space, $a^* = -bK_S^\dagger$ is the minimum-Euclidean-norm solution. If $\text{rank}(K_S) = r$, every local target defect is reachable to second order, the closure areas form*

$$(51) \quad a^* + \ker_L(K_S), \quad \ker_L(K_S) := \{a \in \mathbb{R}^L : aK_S = 0\},$$

and this affine set has dimension $L - r$ and codimension r . Full second-order reachability implies

$$(52) \quad r \leq \binom{\rho}{2} \leq \binom{m}{2}, \quad \rho \geq \rho^*(r) := \left\lceil \frac{1 + \sqrt{1 + 8r}}{2} \right\rceil.$$

A p -coordinate loop library with area map $a = \eta P$, $P \in \mathbb{R}^{p \times L}$, has effective curvature PK_S and requires $p \geq r$ for full target reachability. When K_S has rank r and $p \geq r$, the matrices P for which PK_S has rank r form an open dense subset of $\mathbb{R}^{p \times L}$.

The bound in (52) counts oriented two-dimensional faces of the attainable phase distribution. A displayed phase plane contributes a face only when policy can move the report in both directions. One small loop moves three target-response horizons along a single curvature vector. To reach an arbitrary three-vector, the design needs at least three independent attainable faces, first available at $\rho = 3$, or a library that combines curvature and Lie-word vectors from several base points.

Transversality of the curvature rank. Let Θ be a finite-dimensional smooth manifold, let $\theta \in \Theta$ index policy technologies, implementation metrics, or phase reports, and let the C^1 map $\mathcal{K} : \Theta \rightarrow \mathbb{R}^{L \times r}$ send a design to its target-curvature matrix. Write $\Sigma_d = \{K : \text{rank}(K) = d\}$.

THEOREM 8 (Rank strata of policy curvature). *Suppose $L \geq r$ and $\mathcal{K}$ is transverse to every matrix-rank stratum Σ_d . For each $d < r$, the set $\mathcal{K}^{-1}(\Sigma_d)$ is either empty or a submanifold of codimension*

$$(53) \quad (L - d)(r - d).$$

Full-rank designs form an open dense subset of Θ . If $L < r$, every target-curvature matrix has rank at most L .

Equation (53) states the regularity condition used by the phase-dimension result. It also shows which design perturbations can move a singular curvature matrix away from its rank stratum. Restrictions on G , q , or R may confine $\mathcal{K}(\Theta)$ to a lower-rank family, in which case transversality fails in a way that can be traced to those economic restrictions.

The oriented-area bridge. Implement a_j by an elementary rectangle in the plane E_j , with orientation $\text{sign}(a_j)$ and side lengths proportional to $\sqrt{|a_j|}$. Concatenating the L rectangles returns the phase point to x . Let the resulting target displacement at the terminal state of the baseline cycle be

$$(54) \quad F_s(a) = aK_s + \mathcal{R}_s(a), \quad \|\mathcal{R}_s(a)\| \leq C\|a\|^{3/2}, \quad C \geq 0$$

on a neighborhood of zero.

THEOREM 9 (Curvature bridge and residual sequence). *Suppose K_s has rank r and (54) holds. For a baseline defect b , let $a^* = -bK_s^\dagger$ and suppose a^* lies in the remainder neighborhood. The bridged defect satisfies*

$$(55) \quad \|b + F_s(a^*)\| \leq C\|K_s^\dagger\|^{3/2}\|b\|^{3/2}.$$

Consider repeated updates at the successive terminal states, suppose every proposed area update lies in its local remainder neighborhood, and suppose the remainder constant is bounded by C and $\|K_s^\dagger\| \leq \beta$ along the sequence. With e_k denoting target-defect size and $M = C\beta^{3/2}$,

$$(56) \quad e_{k+1} \leq Me_k^{3/2}.$$

If $M = 0$, then $e_k = 0$ for every $k \geq 1$. If $M > 0$ and $M^2 e_0 < 1$, then

$$(57) \quad e_k \leq M^{-2}(M^2 e_0)^{(3/2)^k}.$$

The calculation uses the target-curvature matrix, its pseudoinverse, and elementary phase loops. The full state transition enters evaluations of target defects and curvature; matrix inversion takes place in dimensions L and r . Equation (57) supplies a stopping certificate for the repeated bridge.

PROPOSITION 8 (Estimated bridge residual). *Suppose $L \geq r$, $\rho, \delta \geq 0$, $\|\widehat{K} - K\|_{\text{op}} \leq \rho$, $\|\widehat{b} - b\| \leq \delta$, and $\widehat{\sigma}_r > \rho$. Set $\widehat{a} = -\widehat{b}\widehat{K}^\dagger$. Then K and $\widehat{K}$ have target rank r , and*

$$(58) \quad \|b + \widehat{a}K\| \leq \delta + \frac{\rho}{\widehat{\sigma}_r} \|\widehat{b}\|.$$

If $\widehat{a}$ lies in the remainder neighborhood, adding the nonlinear term in (54) gives the implementable bound

$$(59) \quad \|b + F_s(\widehat{a})\| \leq \delta + \frac{\rho}{\widehat{\sigma}_r} \|\widehat{b}\| + C\|\widehat{a}\|^{3/2}.$$

4. Dynamic Gorman aggregation

The loop library determines which internal coordinates must accompany the report. Transported target covectors give the local dimension of these coordinates. A finite affine quotient then propagates them across sequences of loops.

4.1. Target-sensitive measurements

Fix a phase loop and linearize its observed history and future probe response around a baseline state. Write $\mathbb{K} = \mathbb{R}$, write perturbations as row vectors $z \in \mathbb{K}^N$, and let

$$(60) \quad z \mapsto zO \in \mathbb{K}^d, \quad z \mapsto zD \in \mathbb{K}^r.$$

The tangent space to the observational fiber is $\{z : zO = 0\}$. The probe response is identified on this fiber when $zO = 0$ implies $zD = 0$.

Let the rows of N_O span $\{z : zO = 0\}$ and set

$$(61) \quad A = N_OD, \quad d^* = \text{rank}(A).$$

Choose a full-row-rank matrix $B \in \mathbb{K}^{d^* \times r}$ whose rows form a basis of $\text{row}(A)$ and a generalized inverse $B^\dagger$ satisfying $zB^\dagger B = z$ on $\text{row}(B)$. Define the target-sensitive measurement matrix

$$(62) \quad M^* = DB^\dagger \in \mathbb{K}^{N \times d^*}.$$

When $d^* = 0$, the measurement and decoder are empty, and the formulas continue to hold with zero-dimensional matrices.

THEOREM 10 (Measurement dimension on an observational fiber). *For every z satisfying $zO = 0$,*

$$(63) \quad zD = (zM^*)B.$$

If a d_M -coordinate measurement matrix M admits a linear decoder C on the fiber, so that $(N_OM)C = N_OD$, then $d_M \geq d^$. Thus d^* is the minimum dimension among linearly decodable measurements, and (62) attains it.*

For a finite distribution μ , each coordinate of μM^* is a weighted micro moment. If two terminal distributions share the phase report and these d^* moments, equation (63) assigns them the same linearized probe response. Sample averages of the weights estimate the

moments. The resulting empirical object has dimension d^* , even when the distribution has many support points.

The randomized loop experiment uses probe outcomes to identify target curvature through (32). The measurement theorem addresses a related practical problem: a full probe experiment may be unavailable at every policy iteration. In that case, target-sensitive moments can predict and monitor the defect. A confidence region for (O, D) produces a set of candidate measurement ranks, and the operator ball in (47) carries this uncertainty into the area bridge.

4.2. A phase-conditional Gorman form

Gorman's polar form writes household expenditure as

$$(64) \quad e_i(p, u_i) = a_i(p) + b(p)u_i,$$

where the price index b is common across households (Gorman 1953, 1961). Shephard's lemma gives the aggregate Hicksian demand

$$(65) \quad H(p, \mathbf{u}) = \nabla_p A(p) + U \nabla_p b(p), \quad A(p) = \sum_i a_i(p), \quad U = \sum_i u_i.$$

Thus a vector of demands depends on the utility profile through one additive coordinate and a common price-dependent slope.

For the policy problem, let $M : \mathcal{S} \rightarrow \mathbb{R}^g$ collect target-sensitive additive moments and let $B : \mathcal{X} \rightarrow \mathbb{R}^{g \times r}$. A phase-conditional polar form is

$$(66) \quad \tau(s) = \alpha(q(s)) + M(s)B(q(s)).$$

For a finite distribution, $M(\mu) = \mu W$ for a matrix of microstate weights W . Equation (65) is the case $q = p$, $M = U$, $\alpha = \nabla_p A$, and $B = \nabla_p b$.

THEOREM 11 (Polar factorization of target holonomy). *Suppose (66) holds on a regularity domain. For every feasible phase loop γ based at $x = q(s)$ with $s \in \mathcal{D}_\gamma$,*

$$(67) \quad h_\tau(\gamma, s) = [M(\text{Hol}_\gamma(s)) - M(s)]B(x).$$

For $v, w \in \mathcal{E}_x$ on an involutive accessible leaf, define $\kappa_M(s; v, w) = DM_s \Omega_s(v, w)$. Then

$$(68) \quad \kappa_\tau(s; v, w) = \kappa_M(s; v, w)B(x), \quad K_s = K_{M,s}B(x),$$

and $\text{rank}(K_s) \leq \min\{g, \text{rank } K_{M,s}, \text{rank } B(x)\}$.

Every target defect generated inside the phase fiber lies in the row space of $B(x)$. With

$g = 1$, an r -coordinate target has a common polar direction. The single area coordinate of a two-dimensional phase diagram can move the target vector along that direction, and exact vector closure requires the baseline defect to lie on the same row. The factorization also reveals two ways that phase sufficiency can arise. Curvature may leave every moment in M fixed, or the common slope may annihilate the induced moment displacement.

4.3. Local dynamic Gorman closure

Fix $s \in \mathcal{S}$, write $x = q(s)$, and let $V_s = \ker Dq_s$ be the tangent space to the observational fiber. Let $\mathcal{L}_x$ contain the empty word and the feasible holonomy words based at x whose maps are defined on a neighborhood of s . For $\gamma \in \mathcal{L}_x$, the vertical differential

$$(69) \quad T_{\gamma,s} = D\text{Hol}_\gamma(s)|_{V_s} : V_s \longrightarrow V_{\text{Hol}_\gamma(s)}$$

transports a hidden-state perturbation around the reported loop. Pull the endpoint target sensitivities back to the initial fiber and define

$$(70) \quad \mathcal{W}_{\tau,s}^* = \text{span} \left\{ T_{\gamma,s}^* \left(D\tau_{\text{Hol}_\gamma(s)}^a |_{V_{\text{Hol}_\gamma(s)}} \right) : \gamma \in \mathcal{L}_x, a = 1, \dots, r \right\} \subseteq V_s^*.$$

By the chain rule, each generator in this span equals $D(\tau^a \circ \text{Hol}_\gamma)_s|_{V_s}$. The loop library, target family, and policy-induced connection jointly determine $\mathcal{W}_{\tau,s}^*$.

For a measurement $M = (M_1, \dots, M_g)$, define first-order loop sufficiency at s by

$$(71) \quad \ker(DM_s|_{V_s}) \subseteq \bigcap_{\gamma \in \mathcal{L}_x} \ker(D(\tau \circ \text{Hol}_\gamma)_s|_{V_s}).$$

When the state is a distribution μ , a Gorman measurement has coordinates $M_j(\mu) = \int \psi_j(z) d\mu(z)$, so each coordinate aggregates a microstate weight by addition.

THEOREM 12 (Local dynamic Gorman closure). *A C^1 measurement $M : \mathcal{S} \rightarrow \mathbb{R}^g$ satisfies (71) at s if and only if*

$$(72) \quad \mathcal{W}_{\tau,s}^* \subseteq \text{span} \{ dM_{1,s}|_{V_s}, \dots, dM_{g,s}|_{V_s} \}.$$

Consequently $\text{rank}(DM_s|_{V_s}) \geq d_G^{\text{loc}}(s)$ for every first-order loop-sufficient measurement, where

$$(73) \quad d_G^{\text{loc}}(s) = \dim \mathcal{W}_{\tau,s}^*.$$

If $\mathcal{W}_{\tau,s}^*$ admits a basis consisting of differentials of additive moment functionals, those $d_G^{\text{loc}}(s)$ functionals attain the bound among additive measurements. Thus additive attainability is equivalent to the existence of an additive differential frame for the transported target span.

For an environment library $\mathcal{J}_0$, write $\Psi_{\eta,\gamma}^a = \tau^{\eta,a} \circ \text{Hol}_\gamma^\eta$, let $T_{\eta,\gamma,s} = D\text{Hol}_\gamma^\eta(s)|_{V_s}$, and

define the shock-level response span

$$(74) \quad \mathcal{W}_{\tau, \mathcal{J}_0, s}^* = \text{span} \left\{ T_{\eta, \gamma, s}^* \left(D\tau_{\text{Hol}_\gamma^\eta(s)}^{\eta, a} |_{V_{\text{Hol}_\gamma^\eta(s)}} \right) : \eta \in \mathcal{J}_0, \gamma \in \mathcal{L}_x^\eta, a = 1, \dots, r \right\}.$$

Fix $\eta_0 \in \mathcal{J}_0$ in the interior of the environment domain and a loop library feasible on its neighborhood. The shock-derivative span and the span required to preserve both the prespecified shock family and the first shock derivative are

$$(75) \quad \mathcal{J}_{\tau, \eta_0, s}^* = \text{span} \left\{ \partial_\eta D_s \Psi_{\eta, \gamma}^a(s) \big|_{\eta=\eta_0} : \gamma \in \mathcal{L}_x, a = 1, \dots, r \right\}, \quad \mathcal{W}_{\tau, \mathcal{J}_0, \eta_0, s}^{*, 1} = \mathcal{W}_{\tau, \mathcal{J}_0, s}^* + \mathcal{J}_{\tau, \eta_0, s}^*.$$

Here level sufficiency preserves first-order state variation of every $\Psi_{\eta, \gamma}$ at the prespecified values of η . Shock-derivative robustness additionally preserves first-order state variation of $\partial_\eta \Psi_{\eta, \gamma}$ at η_0 .

COROLLARY 2 (Shock-library Gorman dimension). *A measurement is first-order sufficient for every prespecified environment, target, and loop in (74) exactly when its fiber differentials span $\mathcal{W}_{\tau, \mathcal{J}_0, s}^*$. The minimum differentiable measurement rank is*

$$(76) \quad d_{G, 0}^{\text{loc}}(s; \mathcal{J}_0) = \dim \mathcal{W}_{\tau, \mathcal{J}_0, s}^*,$$

with attainment by additive moments when the joint span admits an additive differential frame. For $\mathcal{J}_1 \subseteq \mathcal{J}_2$,

$$(77) \quad d_{G, 0}^{\text{loc}}(s; \mathcal{J}_1) \leq d_{G, 0}^{\text{loc}}(s; \mathcal{J}_2).$$

The minimum rank for level sufficiency over $\mathcal{J}_0$ together with shock-derivative robustness at η_0 is

$$(78) \quad d_{G, 1}^{\text{loc}}(s; \mathcal{J}_0, \eta_0) = \dim \mathcal{W}_{\tau, \mathcal{J}_0, \eta_0, s}^{*, 1} \geq d_{G, 0}^{\text{loc}}(s; \mathcal{J}_0),$$

and equality holds exactly when $\mathcal{J}_{\tau, \eta_0, s}^ \subseteq \mathcal{W}_{\tau, \mathcal{J}_0, s}^*$. Additive moments attain the joint bound when the enlarged span admits an additive differential frame.*

The theorem and corollary translate the loop library into a local measurement rule. Each basis covector assigns weights to microstate perturbations, and its pullback incorporates every policy history and environment included in the library. Enlarging the target, shock, or loop library can increase the dimension needed for level sufficiency; the shock jet may add further coordinates. A common C^1 decoder over an open environment neighborhood implies jet closure by differentiation. Closure verified only on a finite shock grid covers levels at those grid points, leaving the derivative between them to be studied separately. If the baseline s_η° moves with the environment, the aggregate comparative static also includes the observed tangent $D(q, M)_{s_\eta^\circ} \partial_\eta s_\eta^\circ$. Level and jet closure recover its

target effect from this tangent. These statements concern first-order variation at s . The next result integrates a constant-rank family into a local quotient, followed by an exact finite affine construction.

4.4. The dynamic target quotient

Fix a phase point x and a relative-open subset U of its fiber that is invariant under a finite library of feasible holonomy maps $g_1, \dots, g_J$. Let $\mathcal{L}_x$ contain the empty word and every finite word in this library. For $\gamma \in \mathcal{L}_x$, write g_γ for the corresponding composition and $\Psi_\gamma = \tau \circ g_\gamma$. Two states have the same dynamic target type when

$$(79) \quad s \equiv_{\tau, \mathcal{L}_x} s' \iff \Psi_\gamma(s) = \Psi_\gamma(s') \text{ for every } \gamma \in \mathcal{L}_x.$$

The relation is invariant under each generator: $s \equiv_{\tau, \mathcal{L}_x} s'$ implies $g_j(s) \equiv_{\tau, \mathcal{L}_x} g_j(s')$. A C^1 representation $M : U \rightarrow \mathbb{R}^g$ is dynamically target-sufficient when every Ψ_γ has a C^1 decoder through M . It is transport-closed when each $M \circ g_j$ has a C^1 update through M .

On the fiber, the transported target span in (70) becomes the response codistribution

$$(80) \quad \mathcal{W}_{\tau, s}^* = \text{span} \{ D\Psi_\gamma^a(s) : \gamma \in \mathcal{L}_x, a = 1, \dots, r \} \subseteq T_s^* U,$$

where Ψ_γ^a is target coordinate a after word γ .

THEOREM 13 (Dynamic target quotient under policy holonomy). *Suppose $U_0 \subseteq U$ is a generator-invariant coordinate neighborhood and $\mathcal{W}_\tau^*$ is a smooth codistribution of constant rank d on U_0 . Suppose d scalar word-response coordinates have differentials that form a frame for $\mathcal{W}_\tau^*$ on U_0 . Let $F : U_0 \rightarrow \mathbb{R}^d$ be their joint map and suppose every nonempty level set $F^{-1}(y)$ in U_0 is connected. Then there is a C^1 submersion $M^* : U_0 \rightarrow \mathbb{R}^d$ whose components are these response coordinates and whose fibers are the local equivalence classes in (79). The representation M^* is dynamically target-sufficient and transport-closed. Every dynamically target-sufficient C^1 representation M satisfies*

$$(81) \quad \text{rank } DM_s \geq d \quad \text{for every } s \in U_0.$$

Thus $d = \text{rank } \mathcal{W}_\tau^$ is the minimum local differentiable state dimension that propagates the target through every word in the loop library.*

The quotient records state propagation separately from the condition for cycle closure. Its induced update describes how a closed report path moves the target-relevant aggregate state. A loop closes the target at s when the decoder has the same value at $M^*(s)$ and at the updated quotient state. For a small oriented loop, curvature is the local derivative of this update.

4.5. Holonomy-invariant polar quotients

The polar form in (66) decodes a target at one state. Prediction after loop concatenation also requires an evolution law for the moments. Consider a finite distribution and let $H \in \mathbb{K}^{N \times h}$ contain the constant and phase-known loadings. Let $P_1, \dots, P_J$ be a library of linear policy holonomies that map the relevant distribution domain into itself and satisfy $P_j H = H$, so every map preserves the phase report μH . A g -coordinate polar quotient consists of W , B , C , and matrices (A_j, R_j) satisfying

$$(82) \quad D = HC + WB, \quad P_j W = HA_j + WR_j, \quad j = 1, \dots, J.$$

Writing $h_x = \mu H$ and $m = \mu W$, these equations give

$$(83) \quad \tau(\mu) = h_x C + mB, \quad m(\mu P_j) = h_x A_j + mR_j.$$

The first equation is a Gorman-form target aggregator. The second makes the additive coordinates an affine state representation of policy holonomy.

Let $\pi : \mathbb{K}^N \rightarrow \mathbb{K}^N / \text{col}(H)$ be the quotient map and let $\bar{P}_j$ denote the map induced by P_j . Define

$$(84) \quad \bar{V}_\infty = \text{span} \left\{ \bar{P}_{i_k} \cdots \bar{P}_{i_1} \pi(Dc) : k \geq 0, i_\ell \in \{1, \dots, J\}, c \in \mathbb{K}^r \right\}, \quad d_G = \dim \bar{V}_\infty.$$

THEOREM 14 (Holonomy-polar aggregation). *The minimum number of additive coordinates in (82) equals d_G . A quotient attaining d_G is obtained by lifting a basis of $\bar{V}_\infty$ to the columns of W . The iteration*

$$(85) \quad \bar{V}_0 = \text{span} \pi(\text{col } D), \quad \bar{V}_{\ell+1} = \bar{V}_\ell + \sum_{j=1}^J \bar{P}_j \bar{V}_\ell$$

reaches $\bar{V}_\infty$ after at most $N - \text{rank}(H)$ strict enlargements.

COROLLARY 3 (Additive realization of the dynamic target quotient). *In the finite linear setting of theorem 14, let U be a relative-open subset of the relative interior of a phase fiber, so its tangent space is $\mathcal{T}_H = \{z \in \mathbb{K}^N : zH = 0\}$. Let $g_j(\mu) = \mu P_j$ and let $\tau(\mu) = \mu D$. The response codistribution in (80), restricted to $\mathcal{T}_H$, has rank d_G . The matrix W constructed in theorem 14 realizes $M^*(\mu) = \mu W$ with additive microstate coordinates. Every linear additive representation that propagates the target through the loop library has at least d_G coordinates.*

The target kernel and policy-loop library therefore produce a finite subspace algorithm. Exact linear aggregation has $d_G = 1$ when a common Gorman coordinate spans the generated subspace. Larger values count the additional additive statistics needed to carry the target through policy history. This construction is related to finite-state lumpability

(Buchholz 1994). Here a specified counterfactual kernel generates the quotient, and the transition maps are policy holonomies based on the phase diagram.

COROLLARY 4 (Finite-cycle polar closure). *For one loop with $m^+ = h_x A + mR$, the k -cycle moment and target defects are*

$$(86) \quad m_k = m_0 R^k + h_x A \sum_{\ell=0}^{k-1} R^\ell, \quad \tau(\mu P^k) - \tau(\mu) = \left[m_0 (R^k - I_g) + h_x A \sum_{\ell=0}^{k-1} R^\ell \right] B.$$

The target closes after k cycles exactly when the bracketed vector belongs to the right kernel of B . The two identities $R^k = I_g$ and $A \sum_{\ell=0}^{k-1} R^\ell = 0$ give state-independent closure on the polar quotient.

Equation (86) gives the economic counterpart of finite-cycle closure for a trajectoid. The phase point returns after every traversal, while the affine matrix R transports the hidden aggregate orientation. The accumulated translation determines the cycle count at which the target also returns.

4.6. Finite-loop polar transport inference

Let $\gamma = (j_1, \dots, j_T)$ be a word in the loop library, with every loop based at the same phase point. Define

$$(87) \quad R_\gamma = R_{j_1} \cdots R_{j_T}, \quad a_\gamma = \sum_{t=1}^T A_{j_t} R_{j_{t+1}} \cdots R_{j_T},$$

where the empty matrix product is I_g . Repeated use of (83) gives

$$(88) \quad m_T = m_0 R_\gamma + h_x a_\gamma, \quad h_\tau(\gamma) = [m_0 (R_\gamma - I_g) + h_x a_\gamma] B.$$

The coefficients can be estimated from target-sensitive moments without observing the full distribution vector. For loop type j , let episode i supply the phase-known row h_{ij} , the target-sensitive moments m_{ij} and m_{ij}^+ before and after the loop, and a probe target τ_{ij} . Write $z_{ij} = (h_{ij}, m_{ij})$ and

$$(89) \quad m_{ij}^+ = z_{ij} \Theta_j + \eta_{ij}, \quad \Theta_j = \begin{bmatrix} A_j \\ R_j \end{bmatrix}, \quad \tau_{ij} = h_{ij} C + m_{ij} B + \nu_{ij}.$$

THEOREM 15 (Polar transport identification and inference). *Let n denote the number of independently sampled episode clusters. Suppose the regression errors in (89) have conditional mean zero and the cluster score vectors have uniformly bounded $2 + \delta$ moments. Suppose every loop type has a limiting episode share in $(0, 1)$, the loop-type sample second moments converge*

in probability to positive-definite matrices $Q_j = \mathbb{E}[z'_{ij}; z_{ij} \mid j]$, and the target-regression sample second moment converges to its positive-definite population counterpart. Suppose the stacked cluster scores satisfy the Lindeberg condition and their covariance matrices converge to a finite matrix Ω . Any estimated initial moments entering the plug-in target are assumed to have a joint asymptotically linear representation in the same cluster array; fixed initial moments satisfy this condition with a zero influence term. Then $(A_j, R_j)_{j=1}^J$, B , and C are identified by the population regressions, and their least-squares estimators are jointly asymptotically linear as $n \rightarrow \infty$. For every fixed loop word γ , the plug-in estimator obtained from (88) satisfies

$$(90) \quad \sqrt{n} \left(\widehat{h}_\tau(\gamma) - h_\tau(\gamma) \right) \implies \mathcal{N}(0, J_\gamma \Sigma J'_\gamma),$$

where Σ is the joint cluster covariance of the coefficient and initial-moment estimators and J_γ is the Jacobian of (88). An estimated or cross-fitted measurement matrix W enters (90) when its estimator has a joint asymptotically linear representation with an $o_p(n^{-1/2})$ remainder; its influence term is appended to Σ .

The transport estimator targets the finite-loop defect in (88). Its normalization uses matrix products and a common slope. The direct curvature estimator instead scales a randomized arm contrast by ε^2 . Direction reversal identifies the local geometric derivative, while polar transport predicts finite policy cycles from a loop-stable aggregate state law. Comparing their implications across feasible loop sizes provides a specification check for the quotient.

5. Evidence and implementation

Two applications show how the objects above can be measured and implemented. Express records reveal whether additive internal coordinates improve predictive closure and provide scale estimates for an assigned-loop experiment. A transition economy then computes policy transport and finite-loop defects within a forward-looking equilibrium.

5.1. Express operations

Operational reports and additive internal state. An Express network contains parcels of different ages and locations, couriers carrying different loads, and delivery areas with different task concentrations. Headquarters may summarize this state by reporting throughput, mean lead time, total backlog, and active capacity. Each operational region is treated as a high-dimensional organization, and the empirical question is whether this report is sufficient to predict its subsequent operating state. The public LaDe release records package acceptance and delivery events from operations in five Chinese cities (Wu et al. 2025).

For region r and day t , define the reported state

$$(91) \quad q_{rt} = (\log(1 + D_{rt}), \bar{\ell}_{rt}, \log(1 + B_{rt}), \log(1 + C_{rt})),$$

where D_{rt} is delivered parcels, $\bar{\ell}_{rt}$ is mean lead time among deliveries with lead time at most seven days, B_{rt} is end-of-day backlog, and C_{rt} is the number of active couriers. The reported lead-time coordinate is set to zero on days without a measured delivery; the throughput coordinate records that event.

The Gorman coordinates are sums over parcels, couriers, or delivery areas. Let a_{it} be the age of backlogged parcel i , let L_{crt} and F_{crt} be courier c 's backlog and delivered tasks, let N_{art} be delivered tasks in area of interest a , and let ℓ_{it} be parcel lead time. The six coordinates are

$$(92) \quad M_{rt} = \left(\sum_{i \in B_{rt}} a_{it}, \sum_{i \in B_{rt}} a_{it}^2, \sum_c L_{crt}^2, \sum_c F_{crt}^2, \sum_a N_{art}^2, \sum_{i \in D_{rt}} \ell_{it}^2 \right).$$

The first two coordinates retain the backlog-age distribution. The next three describe concentration across couriers and delivery areas conditional on reported totals, and the last records delivery-time dispersion. Each coordinate has the additive form in Theorem 12: it sums a micro-unit weight over parcels, couriers, or areas. The target family contains future log throughput, mean lead time, and log backlog at horizons of one, three, and seven days. Backlog obeys the accounting relation $B_{r,t+h} = B_{rt} + \sum_{j=1}^h A_{r,t+j} - \sum_{j=1}^h D_{r,t+j}$. Conditional on a common future arrival path, the target therefore measures cumulative service capacity as a function of the initial internal state. Let H_{rt} collect the current report, its observed history, calendar information, and accepted-parcel flow. The following identity expresses the Gorman closure condition as a predictive-risk estimand.

PROPOSITION 9 (Predictive closure risk). *For a square-integrable scalar target Y , let $\mu_H = \mathbb{E}[Y | H]$ and $\mu_{HM} = \mathbb{E}[Y | H, M]$. The oracle squared-risk gain from carrying the additive state with the report is*

$$(93) \quad \inf_{g \in L^2(H)} \mathbb{E}[(Y - g)^2] - \inf_{f \in L^2(H, M)} \mathbb{E}[(Y - f)^2] = \mathbb{E}[(\mu_{HM} - \mu_H)^2].$$

The report history is conditionally mean-sufficient for Y exactly when the right side is zero.

The sample analysis compares fixed learners on dates excluded from outcome fitting. The squared-loss contrast estimates a learner- and split-specific version of equation (93). If both learners consistently estimate their conditional means, this contrast converges to the oracle quantity. Absolute loss is also reported using the same fixed predictions.

TABLE 1. Predictive closure of the Express report

Target	Horizon	History-flow RMSE	History-flow + moments RMSE	Change (%)
Backlog	1	0.3945	0.3703	6.13
Backlog	3	0.4810	0.4846	-0.74
Backlog	7	0.7111	0.6795	4.44
Lead time	1	10.4928	12.1678	-15.96
Lead time	3	14.9590	15.0482	-0.60
Lead time	7	16.1213	16.1444	-0.14
Throughput	1	0.6074	0.6115	-0.69
Throughput	3	1.0041	1.0063	-0.22
Throughput	7	1.3959	1.3859	0.72

The benchmark contains the current report, current accepted flow, and exact-calendar lags at one, two, three, and seven days. The moment model adds equation (92) and current-report-moment interactions. Change is the percentage reduction in RMSE. Backlog and throughput targets are in logs; lead-time targets are in hours.

Construction and predictive-closure design. After event validation, the sample contains 4,514,658 package records, 4,881 couriers, 172 operational regions, and 30,534 region-days. The analysis uses the 166 regions observed for at least 60 days. Online Appendix H describes the event accounting and variable construction.

For each target $\tau_{r,t+h}$, the analysis compares three specifications. The first uses the contemporaneous report. The history-flow benchmark adds accepted parcels and exact-calendar lags of reports and flows. The polar model further adds $M_{rt}B_h(q_{rt})$ with report-dependent common slopes. Within each region, the first 80 percent of dates with an observed target form the estimation sample, and the remaining dates form the evaluation sample. Sequential validation windows select penalties. Evaluation uses squared and absolute prediction loss with two-way cluster covariance by operational region and calendar date.

The second design forms pairs matched on report and flow history. Within each city-date cell and target sample, a greedy assignment creates disjoint pairs that minimize Euclidean distance in a standardized 25-coordinate vector. The vector contains the current report, current accepted flow, and four exact-calendar lags of each. For every pair, the observed future-outcome difference is compared with the differences predicted by the history-flow model and the Gorman correction. This is the operational analogue of a fiber comparison: the two regional units have nearby reported histories, the benchmark absorbs region and calendar terms, and their additive internal coordinates may differ.

For next-day log backlog, exact-calendar report history and accepted flow reduce RMSE from 0.4228 to 0.3945, and the six additive moments reduce it to 0.3703. The final RMSE increment is 6.13 percent. Squared risk falls 11.89 percent, with a region-and-date two-way-clustered loss-gain t statistic of 2.94 and $p = 0.0038$; the Holm-adjusted p value

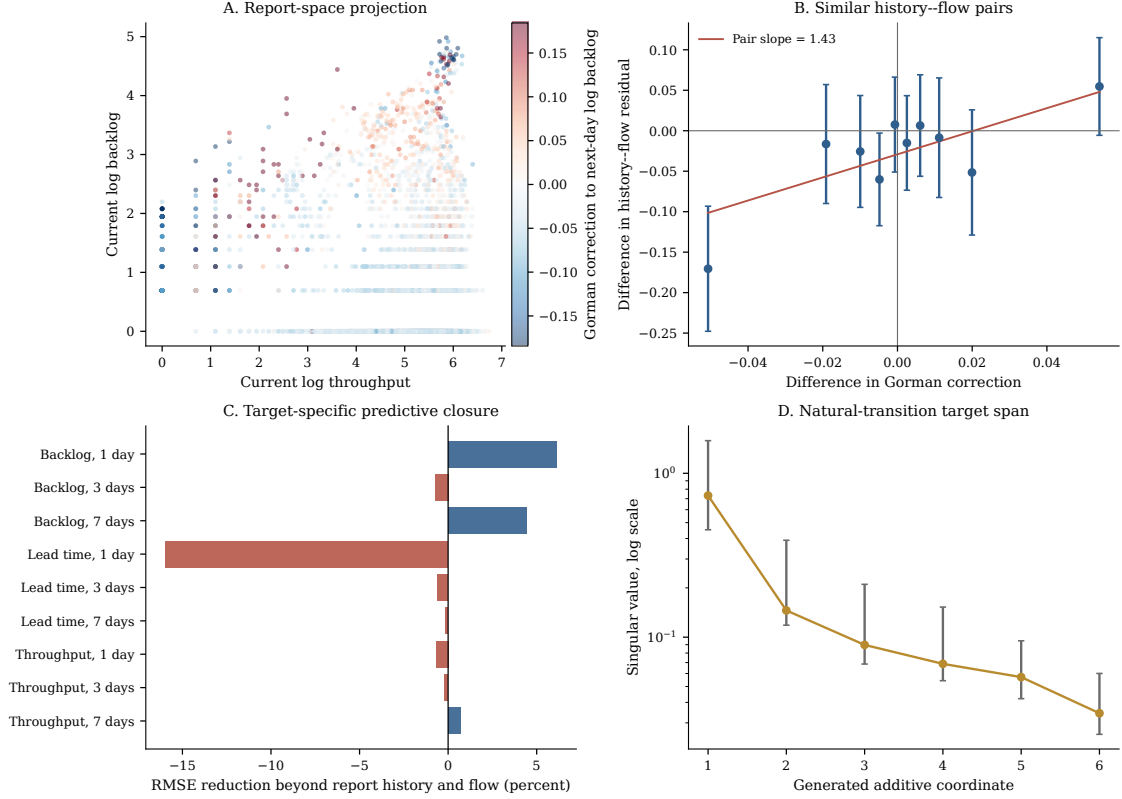


FIGURE 2. Reported operating state and additive internal coordinates

Panel A projects the four-coordinate current report onto log throughput and log backlog for 5,000 held-out region-days; color is the training-estimated Gorman correction to next-day log backlog conditional on report history and accepted flow. Panel B groups the closest half of city-date history-flow-matched pairs into deciles of their Gorman-correction difference; points give mean differences in the benchmark target residual and bars give pointwise 95-percent intervals. Panel C reports out-of-time RMSE reduction from adding the six moments to the history-flow benchmark; positive values indicate lower RMSE. Panel D gives singular values of the additive target span generated by the observed one-day transition; bars are operational-region cluster-bootstrap 95-percent intervals conditional on report history and accepted flow. The LaDe records contain observed operating transitions.

within the nine-target squared-loss family is 0.0307. Absolute risk falls 8.83 percent, with $t = 6.17$ and an eighteen-comparison Holm-adjusted p value of 1.25×10^{-7} . Backlog-age, courier-distribution, and delivery-area/service blocks reduce benchmark RMSE by 5.59, 6.75, and 6.43 percent, respectively. Seven-day backlog RMSE falls 4.44 percent and absolute risk falls 7.18 percent, while the three-day measures change by -0.74 and -0.78 percent. Throughput RMSE changes are -0.69 , -0.22 , and 0.72 percent; lead-time changes are -15.96 , -0.60 , and -0.14 percent. The variation across outcomes and horizons follows the target dependence of $\mathcal{W}_{\tau,s}^*$.

Figure 3 reports checks across time, workload support, location, and future inputs. The next-day gain remains positive in rolling-origin windows, under workload restrictions, in all five cities, and in leave-one-region calculations. In a conditional-alignment exercise,

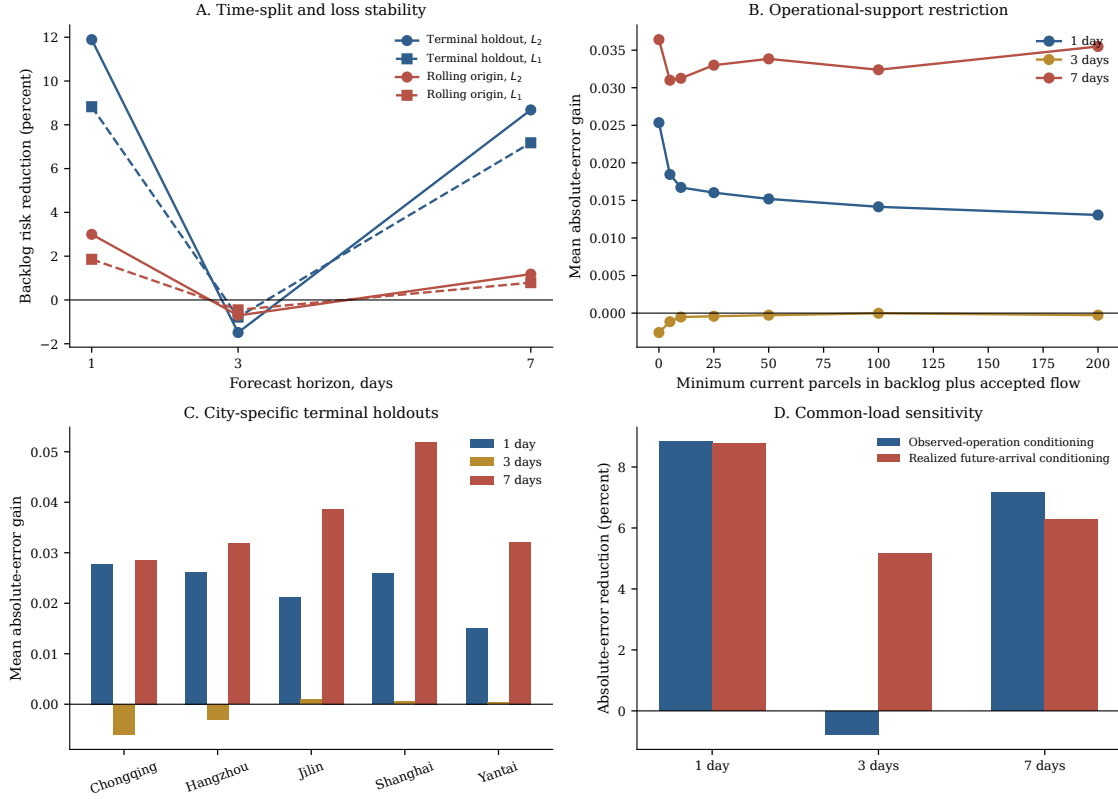


FIGURE 3. Predictive-risk stability across time, support, location, and future load

Panel A gives percentage backlog-risk reductions on the terminal holdout and four disjoint rolling-origin holdouts under squared and absolute loss. Panel B restricts the terminal holdout by minimum current workload and plots absolute-loss gains. Panel C gives city-specific absolute-loss gains. Panel D compares the observed-operation conditioning set with a specification that gives both learners realized cumulative future arrivals through the target horizon. Positive values indicate lower loss after adding the six Gorman coordinates.

the observed next-day and seven-day backlog gains exceed all 999 within-cell permutation values under both losses. Providing both learners with realized cumulative future arrivals leaves absolute-risk reductions of 8.79, 5.16, and 6.27 percent at one, three, and seven days. Comparisons with nonlinear learners indicate how much of the public-data result is specific to the phase-conditional affine Gorman form. A schema audit of the five pinned LaDe files finds parcel, courier, location, acceptance, delivery, and date fields, with zero randomized orientation-mirror path cells and zero terminal-probe assignment fields. The LaDe support therefore identifies the reported predictive contrasts and descriptive transition span. Policy holonomy uses the prospective assigned-path and common-probe design described below.

The observed one-day transition provides a descriptive transport operator for the six moments. Iterating the fitted operator on the nine target loadings gives singular values (0.732, 0.146, 0.090, 0.069, 0.057, 0.034). The 99-percent energy rank is four. Across

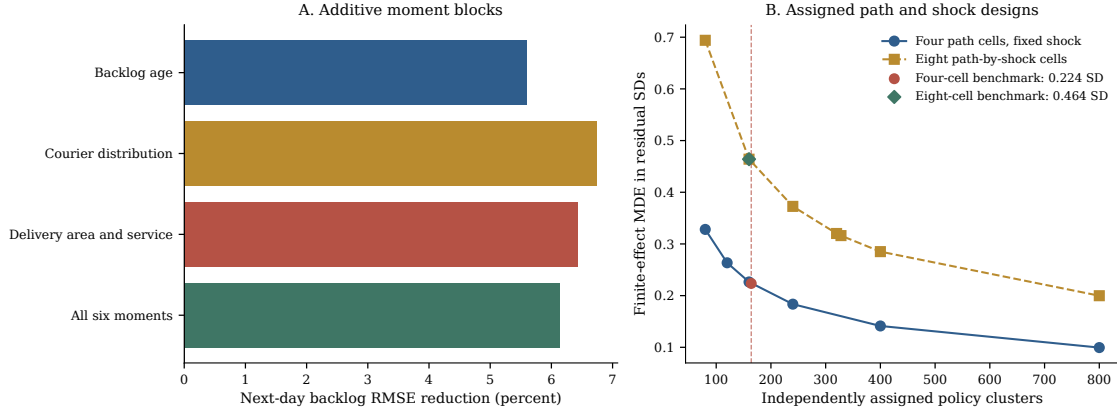


FIGURE 4. Gorman-coordinate blocks and prospective Express assignment

Panel A decomposes the next-day backlog RMSE reduction beyond report history and accepted flow. Each two-coordinate block enters with current-report interactions; the final bar uses all six coordinates. Panel B gives two-sided five-percent, 80-percent-power minimum detectable finite effects in matched-set residual standard deviations. The four-cell curve holds the terminal shock fixed across orientation-by-mirror arms. The eight-cell curve estimates the difference between high-shock and low-shock four-arm contrasts. Division by the shock gap converts this difference into a secant threshold. Markers use the largest balanced designs supported by 166 observed regions. LaDe residuals calibrate outcome dispersion; assigned paths and shocks identify the causal effects.

199 operational-region bootstrap draws, it is three in 28 draws, four in 126, and five in 45. This calculation estimates the target span generated by natural transitions, conditional on report history and accepted flow. Because the LaDe files contain historical operations, the result concerns predictive aggregation under observed transitions. A policy-induced connection and causal holonomy require assigned paths and a common terminal probe.

For prospective assignment, next-day residual dispersion implies a four-arm minimum detectable finite effect of 0.0754 log point, or 0.224 residual standard deviation, when 164 graph-separated regions supply 41 clusters to each arm at two-sided five-percent size and 80-percent power. The eight-cell path-shock design has a 0.1564 log-point finite-difference threshold with 160 balanced clusters. Online Appendix H reports retention, compliance, dependence, and multiplicity sensitivities.

An assigned Express experiment could use routing priority, courier incentives, capacity allocation, and service promises as policy coordinates. Graph-separated operational clusters would enter matched sets of four and receive orientation-by-mirror paths centered on their initial reports. An independently randomized terminal demand, priority, or capacity probe would measure backlog clearance and delivery degradation. A second path-environment design would identify the connection comparative static in equation (38). Path assignment identifies curvature, terminal-probe assignment identifies the response, and prepath environment assignment identifies its comparative static. Online Appendix I reports a constructed marketplace benchmark for the assigned-path estimator and bridge.

5.2. Transition equilibrium

Distribution, prices, and path-space lift. The transition economy has 24 asset nodes and three income states. CRRA utility and logit-smoothed asset choice produce a 72-coordinate distribution μ_t . On the reported transition domain, $K(\mu_t) > 0$, $L(\mu_t) > 0$, $Y(\mu_t) > 0$, and $C_t > 0$. Aggregate assets and labor determine production capital and factor prices:

$$(94) \quad \begin{aligned} K(\mu_t) &= \zeta \mu_t a, & L(\mu_t) &= \mu_t e, & Y(\mu_t) &= ZK(\mu_t)^\alpha L(\mu_t)^{1-\alpha}, \\ w(\mu_t) &= (1 - \alpha) \frac{Y(\mu_t)}{L(\mu_t)}, & r(\mu_t) &= \alpha \frac{Y(\mu_t)}{K(\mu_t)} - \delta. \end{aligned}$$

The calibration uses $\alpha = 0.36$, $\delta = 0.025$, $\zeta = 18.446$, and $Z = 0.57496$, which reproduce a wage of one and a net return of one percent at the stationary distribution. For every policy path, household value functions are solved backward along the endogenous wage and return path, distributions are propagated forward, and market prices are updated to a fixed point. A 12-quarter zero-policy tail joins the loop to the stationary continuation value. This is a finite-horizon perfect-foresight transition equilibrium in the sequence-space tradition (Reiter 2009; Auclert et al. 2021).

The policy vector contains a return wedge, a wealth-tilted transfer, and an income-tilted transfer in every quarter. The report is

$$(95) \quad q_t = 100 \left(\log \frac{Y(\mu_{t+1})}{\bar{Y}}, \log \frac{C_t}{\bar{C}} \right).$$

The path map $\Phi_{12} : \mathbb{R}^{36} \rightarrow \mathbb{R}^{24}$ includes the backward household solution, forward distribution, factor prices, and report sequence. The path-space lift in proposition 2 applies a minimum-norm Newton correction to all 36 policy coordinates. Its computed Jacobian has rank 24 along every reported loop. The smallest singular value across the four scales and two orientations is 0.0386, leaving a 12-dimensional local set of policy paths for each requested report path.

Closed output-consumption paths and future probes. For normalized path time $\theta \in [0, 1]$, orientation $o \in \{-1, 1\}$, and scale s , the requested report loop is

$$(96) \quad \gamma_{o,s}(\theta) = (os(0.85) \sin(2\pi\theta), s(0.45)[1 - \cos(2\pi\theta)]).$$

One traversal lasts 12 quarters. The standardized probe begins with a return-policy coordinate of -1.5 and decays at rate 0.68 over eight quarters. Its ordinary consumption effects at horizons 0, 3, and 7 are $(-0.1183, -0.3934, -0.4414)$ percentage point. Figure 5 plots the scale-0.1 paths and the horizon-0 defects over four scales.

The counterclockwise defect rises from 0.00043 percentage point at scale 0.1 to 0.00714

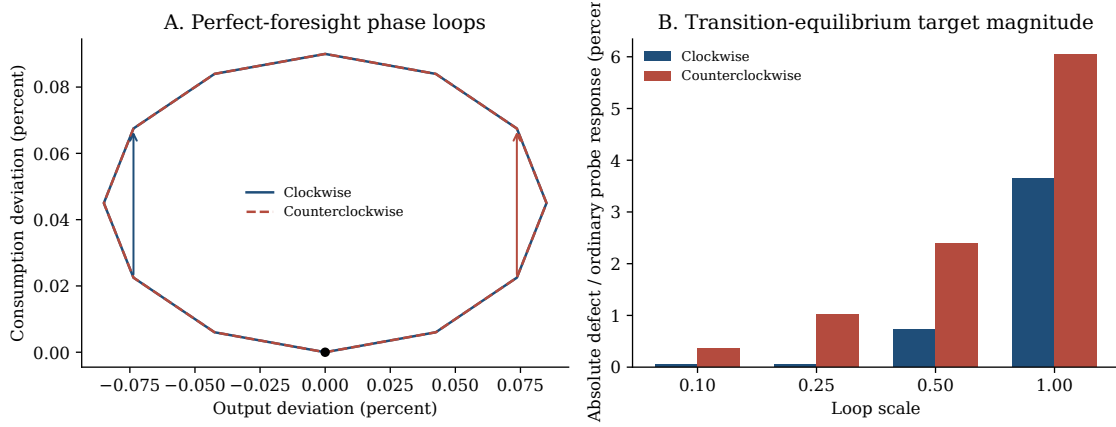


FIGURE 5. Policy holonomy in transition equilibrium

Panel A gives clockwise and counterclockwise 12-quarter output–consumption paths at scale 0.1. Both paths return to the stationary report point. Panel B reports the absolute horizon-0 consumption-probe defect divided by the ordinary horizon-0 probe response. Household decisions, factor prices, and distributions satisfy the perfect-foresight transition equilibrium along each path.

TABLE 2. Transition-equilibrium consumption-probe defects

Scale	CW defect	CCW defect	CW ratio (%)	CCW ratio (%)
0.10	0.00006	-0.00043	0.05	0.36
0.25	-0.00007	-0.00121	0.06	1.02
0.50	-0.00086	-0.00283	0.73	2.39
1.00	-0.00432	-0.00714	3.65	6.04

CW and CCW denote clockwise and counterclockwise. Defects are percentage-point changes in the horizon-0 response to the common probe, relative to the stationary initial distribution. Ratios divide absolute defects by the ordinary response of 0.11826 percentage point.

at scale 1, corresponding to 0.36 and 6.04 percent of the ordinary response. The counterclockwise terminal-distribution distance rises from 0.00334 to 0.03961 in ℓ_1 . The maximum report-path error across the table is 9.53×10^{-6} percentage point. At scale 0.1, the maximum policy-coordinate norm is 0.824 clockwise and 0.628 counterclockwise. This scale sequence records finite path transport in the transition equilibrium. The fixed-price calculation in the online appendix supplies the shrinking-loop curvature and bridge diagnostics, while the current-value general-equilibrium calculation supplies a second finite-loop polar-transport diagnostic.

6. Conclusion

When a policy path returns a reported state to its starting point, part of the economic state can remain displaced. A policy technology, reported drift, and implementation metric

lift attainable report motion to the economic state space, where closed feasible paths act on the fiber above the report. A future policy probe projects that action onto a counterfactual target. Clockwise and counterclockwise paths identify target curvature, and mirrored loops remove the cubic term. Path-space differentiation extends the connection to forward-looking equilibrium. Under partial interference, cluster randomization identifies finite-loop and local contrasts, with policy assignments and terminal micro observations contributing separately to precision.

The same curvature map can be used to design closing paths. Its row space contains the bridgeable defects, its rank determines closure codimension, and attainable report rank determines the available area directions. Elementary loops implement the pseudoinverse area vector. When access is noninvolutive, Lie-word controls provide further directions. Repeated updates produce a residual bound in target dimension, and singular-value bounds incorporate estimation uncertainty in both defects and curvature.

Target-sensitive measurements take a polar form with common slopes. Pulling endpoint target covectors back through the loop library gives the local measurement span and its dimension. An additive differential frame attains the Gorman bound, while constant rank integrates the span into a local target quotient. In the finite linear setting, invariant-subspace iteration constructs the additive coordinates and the finite-word formula propagates them. The transition economy gives a closed output-consumption path that changes a future consumption response by as much as 6.04 percent of its ordinary value. In LaDe Express operations, additive backlog-age and concentration moments reduce next-day and seven-day backlog risk beyond report history and accepted flow. Checks across time, workload, city, conditional alignment, and future arrivals describe where the result appears, while nonlinear learners show its functional-form reach. Differences across targets reflect the probe dependence of aggregation. Macroeconomic phase diagrams and organizational dashboards therefore pose the same measurement question through different observational fibers, and assigned policy loops provide the experiment for studying their holonomy.

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